Probability Rules & Conditional Probability

ANSWER KEY

Basic probability rules

1 A fair six-sided die is rolled. Find the probability the roll is even, and the probability the roll is greater than 4.

$P(\text{even}) = P(2, 4, 6) = \frac{3}{6} = 0.5$. $P(\text{greater than } 4) = P(5, 6) = \frac{2}{6} \approx 0.333$. (Written in words: the chance of rolling even is 0.5; the chance of rolling above 4 is about 0.333.)

2 For events A and B, $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cup B) = 0.6$. Find $P(A \cap B)$ using the addition rule.

Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. So $0.6 = 0.4 + 0.3 - P(A \cap B)$, giving $P(A \cap B) = 0.7 - 0.6 = 0.1$.

Mutually exclusive and independent events

3 Explain the difference between mutually exclusive events and independent events, and give a brief example of each.

Mutually exclusive events cannot happen at the same time (e.g. rolling a 2 and rolling a 5 on the same single die roll — $P(A \cap B) = 0$). Independent events do not affect each other's probability of occurring (e.g. flipping heads on one coin and rolling a 6 on a separate die — knowing one outcome tells you nothing about the other).

4 Two events A and B satisfy $P(A) = 0.5$ and $P(B) = 0.5$, and they are mutually exclusive. Explain why A and B cannot also be independent (unless one has probability 0).

If A and B are independent, $P(A \cap B) = P(A)P(B) = 0.25$. But since they are mutually exclusive, $P(A \cap B) = 0$. Since $0.25 \neq 0$, the two events cannot be both mutually exclusive and independent here — in general, two events with positive probability cannot be both mutually exclusive and independent, since mutual exclusivity means one occurring rules out the other, which is a strong dependence, not independence.

Conditional probability

5 In a class, the probability a student likes math is 0.6, and the probability a student likes both math and science is 0.42. Find the probability a student likes science given that they like math.

$P(\text{science given math}) = \frac{P(\text{math and science})}{P(\text{math})} = \frac{0.42}{0.6} = 0.7$.

6 Use the multiplication rule to find $P(A \cap B)$ if $P(A) = 0.5$ and $P(B \mid A) = 0.4$.

Multiplication rule: $P(A \cap B) = P(A) \times P(B \mid A) = 0.5 \times 0.4 = 0.2$.

Two-way tables and conditional probability

7 A survey of 200 students finds 90 own a car, 120 have a part-time job, and 60 both own a car and have a job. Find the probability a randomly selected student has a job, given that they own a car.

$P(\text{job given car}) = \dfrac{P(\text{car and job})}{P(\text{car})} = \dfrac{60/200}{90/200} = \dfrac{60}{90} = \dfrac{2}{3} \approx 0.667.$

8 Using the same survey data, determine whether owning a car and having a job are independent events, showing your reasoning.

$P(\text{job}) = \dfrac{120}{200} = 0.6$. From the previous problem, $P(\text{job given car}) \approx 0.667$. Since these two probabilities are not equal ($0.667 \neq 0.6$), owning a car and having a job are not independent — knowing a student owns a car changes the probability they have a job.

Tree diagrams and sequential events

9 A factory has two machines: Machine A produces 60% of items with a 2% defect rate, and Machine B produces 40% of items with a 5% defect rate. Find the probability a randomly selected item is defective, using the law of total probability.

$P(\text{defective}) = P(A)P(\text{defective} \mid A) + P(B)P(\text{defective} \mid B) = (0.6)(0.02) + (0.4)(0.05) = 0.012 + 0.02 = 0.032$ or 3.2%.

Applying probability rules to real contexts

10 A weather forecaster states there is a 30% chance of rain on Saturday and a 30% chance of rain on Sunday, and treats the two days as independent. Find the probability it rains on both days, and the probability it rains on at least one of the two days.

Both days: $P(\text{Sat and Sun}) = 0.3 \times 0.3 = 0.09$. At least one day: using the complement, $P(\text{neither day}) = (1 - 0.3)(1 - 0.3) = 0.7 \times 0.7 = 0.49$, so $P(\text{at least one}) = 1 - 0.49 = 0.51$.

The complement rule

11 A box of 50 light bulbs contains 6 defective bulbs. If one bulb is selected at random, find the probability it is NOT defective, using the complement rule.

$P(\text{defective}) = \dfrac{6}{50} = 0.12$. $P(\text{not defective}) = 1 - 0.12 = 0.88$.

12 A student estimates there is a 65% chance of rain at least once during a 3-day trip, assuming each day is independent with the same daily rain probability p. Find p, to three decimal places, using $P(\text{no rain all 3 days}) = (1 - p)^3$.

$P(\text{at least one rainy day}) = 1 - (1 - p)^3 = 0.65$, so $(1 - p)^3 = 0.35$. Taking the cube root: $1 - p = (0.35)^{1/3} \approx 0.705$, giving $p \approx 0.295$.

Combining independence and conditional probability

13 Two lie detector tests are given to a suspect, each correctly detecting deception 90% of the time (independently of each other). If the suspect is actually lying, find the probability both tests correctly detect it.

Since the tests are independent: $P(\text{both correct}) = 0.9 \times 0.9 = 0.81$.

14 A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn without replacement. Find the probability both are red.

$P(\text{first red}) = \frac{5}{8}$. Given the first is red, $P(\text{second red} \mid \text{first red}) = \frac{4}{7}$.
$P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14} \approx 0.357$.

15 At a certain hospital, 40% of patients admitted have condition A, and 25% have condition B. Additionally, 10% of all patients have both conditions A and B. Draw on your understanding of the addition rule to find the probability a randomly selected patient has condition A or condition B (or both), and separately explain why simply adding $P(A) + P(B)$ without adjustment would overcount certain patients.

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.40 + 0.25 - 0.10 = 0.55$. Simply adding $P(A) + P(B) = 0.65$ would double-count the patients who have both conditions (the 10% overlap), since those patients would be counted once within the 40% and again within the 25% — subtracting $P(A \cap B)$ corrects for this double-counting.

16 An airline reports that 85% of its flights depart on time. Among on-time flights, 95% also arrive on time, while among delayed departures, only 40% arrive on time. Using a tree-diagram approach and the law of total probability, find the overall probability that a randomly selected flight arrives on time.

$P(\text{arrives on time}) = P(\text{on-time departure}) \times P(\text{on-time arrival} \mid \text{on-time departure}) + P(\text{delayed departure}$
or about 86.75%.

17 A quality inspector at a factory knows that 3% of all items produced are defective. The inspection process correctly identifies a defective item 98% of the time, but also incorrectly flags a non-defective item as defective 2% of the time (a false positive). If an item is flagged as defective by the inspection process, use the given information and the multiplication rule to find the probability that the item actually IS defective, showing every step of your reasoning (this requires combining the law of total probability with conditional probability, similar to Bayes' theorem).

$P(\text{defective and flagged}) = P(\text{defective}) \times P(\text{flagged} \mid \text{defective}) = (0.03)(0.98) = 0.0294$.
$P(\text{not defective and flagged}) = P(\text{not defective}) \times P(\text{flagged} \mid \text{not defective}) = (0.97)(0.02) = 0.0194$. $P(\text{flagged}) = 0.0294 + 0.0194 = 0.0488$.
$P(\text{defective} \mid \text{flagged}) = \frac{P(\text{defective and flagged})}{P(\text{flagged})} = \frac{0.0294}{0.0488} \approx 0.603$, or about 60.3% — notably less than the 98% accuracy rate might naively suggest, because defects are rare to begin with.

18 A survey of 500 adults finds that 280 exercise regularly, 220 eat a balanced diet, and 150 do both. Using this information, first find the probability a randomly selected adult exercises regularly or eats a balanced diet (or both), and then find the probability an adult eats a balanced diet given that they exercise regularly, showing every step of your work clearly.

$P(\text{exercise}) = \frac{280}{500} = 0.56$. $P(\text{diet}) = \frac{220}{500} = 0.44$. $P(\text{both}) = \frac{150}{500} = 0.30$. Using the addition rule: $P(\text{exercise or diet}) = 0.56 + 0.44 - 0.30 = 0.70$. Using the conditional probability formula: $P(\text{diet} \mid \text{exercise}) = \frac{P(\text{both})}{P(\text{exercise})} = \frac{0.30}{0.56} \approx 0.536$, or about 53.6%.